Binary Search

Goal: find a value in a SORTED array.
Idea: compare with the middle element,
then throw away half of the array.

Each step halves the search space,
so it needs about log2(n) steps.

n = 1,000,000 -> about 20 comparisons.

The algorithm

```
lo = 0, hi = n - 1
while lo <= hi:
    mid = (lo + hi) / 2
    if a[mid] == x: return mid
    if a[mid] < x:  lo = mid + 1
    else:           hi = mid - 1
return -1  (not found)
```

Common bugs

1. Forgetting that the array must be sorted.
2. Using lo < hi and missing the last element.
3. Infinite loops: lo = mid instead of mid + 1.
4. Overflow: (lo + hi) / 2 in fixed-size ints;
   prefer lo + (hi - lo) / 2.

Try highlighting this text in the app!